

MACHINE LEARNING FUNDAMENTALS SERIES

Cross Entropy: Formula Simplification, Loss Comparison, and Backpropagation

Source: StatQuest with Josh Starmer — Neural Networks Parts 6 & 7

1. Cross Entropy: Original to Simplified Formula

Cross Entropy measures how well a predicted probability distribution $\hat{y}$ aligns with the observed true class distribution y .

1.1 General Multi-Class Cross Entropy Formula

In information theory and multi-class classification (M classes), the general Cross Entropy loss for a single sample is:

$$H(y, \hat{y}) = - \sum_{c=1}^M y_c \ln(\hat{y}_c) \quad (1)$$

where $y_c \in \{0, 1\}$ is the one-hot encoded ground truth for class c , and $\hat{y}_c \in (0, 1)$ is the SoftMax predicted probability for class c .

1.2 Simplification in Neural Networks

In supervised classification, every observed data point belongs to exactly one true class c^* . Thus:

- $y_{c^*} = 1$ for the true observed class.
- $y_c = 0$ for all non-observed classes ($c \neq c^*$).

Expanding the summation across all M classes:

$$H(y, \hat{y}) = - \left[\underbrace{1 \cdot \ln(\hat{y}_{\text{observed}})}_{\text{True Observed Class}} + \sum_{c \neq c^*} \underbrace{0 \cdot \ln(\hat{y}_c)}_{=0 \text{ (Vanishes)}} \right] \quad (2)$$

All terms for non-observed classes multiply by zero and vanish completely, leaving the **simplified single-term loss formula**:

$$L_i = -\ln(\hat{y}_{\text{observed}, i}) \quad (3)$$

Interpretation: The Cross Entropy loss for sample i is simply the negative natural logarithm of the predicted SoftMax probability corresponding to the actual observed species.

2. Why Probabilities Cannot Use Sum of Squared Residuals (SSR)

When predicting continuous variables, neural networks use the Sum of Squared Residuals ($\text{SSR} = (y - \hat{y})^2$). However, for bounded probabilities $\hat{y} \in (0, 1]$, SSR performs poorly compared to Cross Entropy.

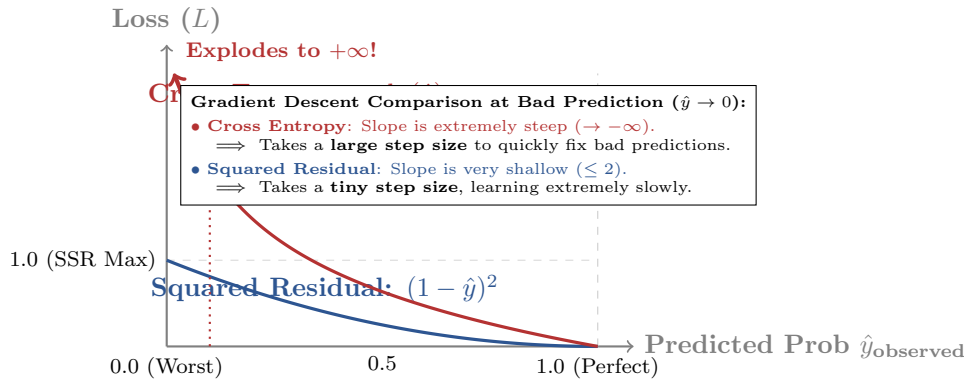


Figure 1: Loss behavior comparison between Cross Entropy and Squared Residuals for probability predictions.

Mathematical Reason for SSR Failure

- Shallow Slopes on Bad Predictions:** For $SSR = (1 - \hat{y})^2$, as $\hat{y} \rightarrow 0$, the loss caps out at 1.0 and the derivative magnitude $|\frac{\partial SSR}{\partial \hat{y}}| = 2(1 - \hat{y})$ is bounded by 2. The network takes tiny steps when it is most wrong.
- Exploding Slopes on Cross Entropy:** For $L = -\ln(\hat{y})$, as $\hat{y} \rightarrow 0$, $L \rightarrow +\infty$. The derivative magnitude $|\frac{\partial L}{\partial \hat{y}}| = \frac{1}{\hat{y}} \rightarrow +\infty$. This provides a massive gradient signal to rapidly correct terrible predictions during Backpropagation.

3. Total Loss and Step-by-Step Derivative for Bias b_3

In Video Part 7, we optimize the output bias b_3 (which directly controls the raw output $z_1 = z_{\text{setosa}}$) across $N = 3$ training samples (Iris Iris dataset: Setosa, Virginica, Versicolor).

3.1 Total Loss Function (Total Cross Entropy)

The Total Loss is the sum of Cross Entropy values across all N training samples:

$$\text{Total Loss} = \sum_{i=1}^N L_i = L_{\text{setosa},1} + L_{\text{virginica},2} + L_{\text{versicolor},3} \quad (4)$$

3.2 Chain Rule Derivative Derivation for Bias b_3

Since b_3 only directly influences z_{setosa} , but z_{setosa} appears in the SoftMax equations for ALL species, we derive two distinct derivative cases via the Chain Rule:

3.2.1 Case A: Observed Species IS Setosa ($i = 1$)

When the actual sample is Setosa, $L_1 = -\ln(\hat{y}_{\text{setosa},1})$. Using the Chain Rule:

$$\frac{\partial L_1}{\partial b_3} = \frac{\partial L_1}{\partial \hat{y}_{\text{setosa}}} \cdot \frac{\partial \hat{y}_{\text{setosa}}}{\partial z_{\text{setosa}}} \cdot \frac{\partial z_{\text{setosa}}}{\partial b_3} \quad (5)$$

- $\frac{\partial(-\ln \hat{y}_{\text{setosa}})}{\partial \hat{y}_{\text{setosa}}} = -\frac{1}{\hat{y}_{\text{setosa}}}$
- $\frac{\partial \hat{y}_{\text{setosa}}}{\partial z_{\text{setosa}}} = \hat{y}_{\text{setosa}}(1 - \hat{y}_{\text{setosa}})$ (SoftMax derivative wrt same class)
- $\frac{\partial z_{\text{setosa}}}{\partial b_3} = 1$ (since $z_{\text{setosa}} = \text{Surface}_{\text{blue}} + \text{Surface}_{\text{orange}} + b_3$)

Multiplying the three terms together:

$$\frac{\partial L_1}{\partial b_3} = \left(-\frac{1}{\hat{y}_{\text{setosa},1}}\right) \cdot (\hat{y}_{\text{setosa},1}(1 - \hat{y}_{\text{setosa},1})) \cdot 1 = \hat{y}_{\text{setosa},1} - 1 \quad (6)$$

3.2.2 Case B: Observed Species IS NOT Setosa ($i = 2, 3$)

When the actual sample is Virginica or Versicolor, $L_i = -\ln(\hat{y}_{\text{other},i})$. Using the Chain Rule:

$$\frac{\partial L_i}{\partial b_3} = \frac{\partial L_i}{\partial \hat{y}_{\text{other}}} \cdot \frac{\partial \hat{y}_{\text{other}}}{\partial z_{\text{setosa}}} \cdot \frac{\partial z_{\text{setosa}}}{\partial b_3} \quad (7)$$

1. $\frac{\partial(-\ln \hat{y}_{\text{other}})}{\partial \hat{y}_{\text{other}}} = -\frac{1}{\hat{y}_{\text{other}}}$ 2. $\frac{\partial \hat{y}_{\text{other}}}{\partial z_{\text{setosa}}} = -\hat{y}_{\text{setosa}} \cdot \hat{y}_{\text{other}}$ (SoftMax derivative wrt different class) 3. $\frac{\partial z_{\text{setosa}}}{\partial b_3} = 1$

Multiplying the three terms together:

$$\frac{\partial L_i}{\partial b_3} = \left(-\frac{1}{\hat{y}_{\text{other},i}}\right) \cdot (-\hat{y}_{\text{setosa},i} \cdot \hat{y}_{\text{other},i}) \cdot 1 = \hat{y}_{\text{setosa},i} \quad (8)$$

3.2.3 Total Gradient Formula for b_3

Combining the partial derivatives across all 3 training samples:

$$\frac{\partial \text{Total Loss}}{\partial b_3} = (\hat{y}_{\text{setosa},1} - 1) + \hat{y}_{\text{setosa},2} + \hat{y}_{\text{setosa},3} \quad (9)$$

4. Step-by-Step Gradient Descent Update for b_3

We optimize b_3 starting from initial value $b_3^{(0)} = -2.0$ with Learning Rate $\eta = 1.0$.

4.1 Iteration 1 Calculation

1. **Forward Pass at $b_3 = -2.0$:**

- Sample 1 (Setosa): $\hat{y}_{\text{setosa},1} = 0.15 \implies \text{Grad}_1 = 0.15 - 1 = -\mathbf{0.85}$.
- Sample 2 (Virginica): $\hat{y}_{\text{setosa},2} = 0.04 \implies \text{Grad}_2 = +\mathbf{0.04}$.
- Sample 3 (Versicolor): $\hat{y}_{\text{setosa},3} = 0.04 \implies \text{Grad}_3 = +\mathbf{0.04}$.

2. **Total Gradient Calculation:**

$$\text{Slope} = -0.85 + 0.04 + 0.04 = -\mathbf{0.77} \quad (\text{Total Error} = 2.67) \quad (10)$$

3. **Step Size Computation ($\eta = 1.0$):**

$$\text{Step Size} = \text{Slope} \cdot \eta = -0.77 \times 1.0 = -\mathbf{0.77} \quad (11)$$

4. **Parameter Update:**

$$b_3^{(1)} = b_3^{(0)} - \text{Step Size} = -2.0 - (-0.77) = -\mathbf{1.23} \quad (12)$$

4.2 Convergence

Repeating the forward pass and gradient update steps shrinks the slope towards zero. The optimization converges at:

$$b_3^* = -\mathbf{0.03} \implies \text{Minimum Total Cross Entropy Loss} = \mathbf{1.72} \quad (13)$$